

Heat Transfer Roadmap

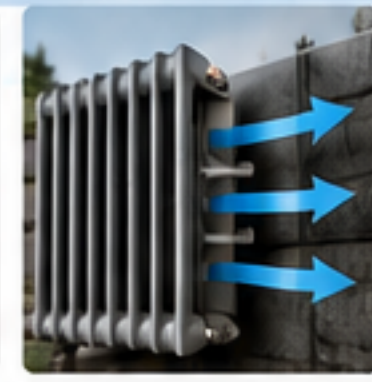
1 FUNDAMENTALS OF HEAT TRANSFER

1.1 Modes of Heat Transfer

- **Conduction** – Heat transfer through a stationary medium due to a temperature gradient.
- **Convection** – Heat transfer between a surface and a moving fluid.
- **Radiation** – Heat transfer through electromagnetic radiation.



Conduction



Convection



Radiation

1.2 Basic Heat Transfer Laws

Fourier's Law of Conduction

$$q_x = -kA \frac{dT}{dx} \quad (1)$$

Where: k – thermal conductivity (W/m K), A – cross-sectional area (m²), dT/dx – temperature gradient (K/m).

Note: The negative sign indicates heat flows in the direction of decreasing temperature.

Newton's Law of Cooling

$$q = hA(T_s - T_\infty) \quad (2)$$

Where: h – convection heat transfer coefficient (W/m²K), T_s – surface temperature, T_∞ – fluid temperature far from surface.



Newton's Law of Cooling
(hot cup cooling in air)

2 CONDUCTION

2.1 One-Dimensional Steady-State Conduction

1. Plane Wall

Temperature distribution (linear for constant k):

$$T(z) = T_1 + (T_2 - T_1) \frac{z}{L} \quad (3)$$

Heat transfer rate and thermal resistance:

$$q = kA \frac{T_1 - T_2}{L}, \quad R_{cond} = \frac{L}{kA} \quad (4)$$



Plane Wall

2. Cylindrical Wall

Temperature distribution (radial conduction):

$$T(r) = T_1 + (T_2 - T_1) \frac{\ln(r/r_1)}{\ln(r_2/r_1)} \quad (5)$$

Heat transfer rate and cylindrical thermal resistance:

$$q = \frac{2\pi kL(T_1 - T_2)}{\ln(r_2/r_1)}, \quad R_{cyl} = \frac{\ln(r_2/r_1)}{2\pi kL} \quad (6)$$



Cylindrical Wall

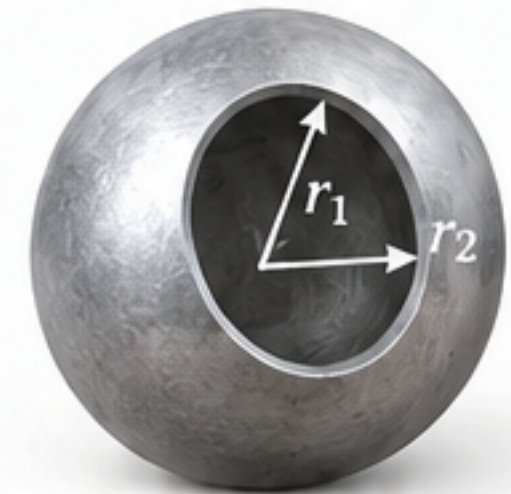
3. Spherical Wall

Temperature distribution:

$$T(r) = T_1 + (T_2 - T_1) \frac{1/r_1 - 1/r}{1/r_1 - 1/r_2} \quad (7)$$

Heat transfer rate and spherical thermal resistance:

$$q = \frac{4\pi k(T_1 - T_2)}{1/r_1 - 1/r_2}, \quad R_{sph} = \frac{1/r_1 - 1/r_2}{4\pi k} \quad (8)$$



Spherical Wall

2.2 Composite Walls

Thermal Resistance Analogy:

Temperature difference → Thermal resistance → Heat transfer rate

Analogs to electrical resistance: $V \leftrightarrow \Delta T$, $I \leftrightarrow q$, $R_{elec} \leftrightarrow R_{th}$.

Series resistance (composite plane wall):

$$q = \frac{T_1 - T_2}{\sum R_i} = \frac{T_1 - T_2}{R_1 + R_2 + \dots + R_n} \quad (9)$$

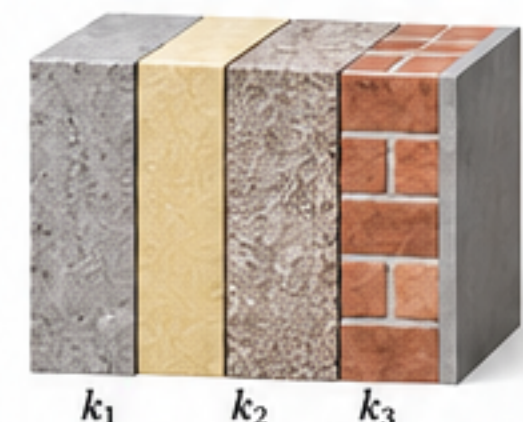
$$R_{total} = \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \dots + \frac{L_n}{k_n A} = \sum \frac{L_i}{k_i A} \quad (10)$$

Parallel resistance:

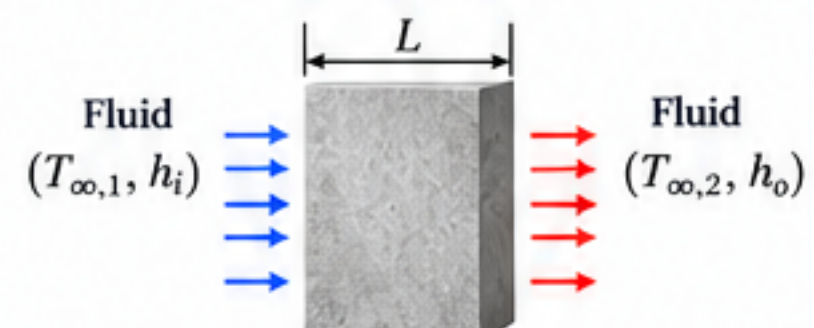
$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n} \quad (11)$$

Overall heat transfer coefficient U (plane wall with convection on both sides):

$$q = UA\Delta T_{overall} = \frac{1}{U} = \frac{1}{h_i} + \frac{L}{k} + \frac{1}{h_o} \quad (12)$$



Composite Plane Wall
(Series Resistance)



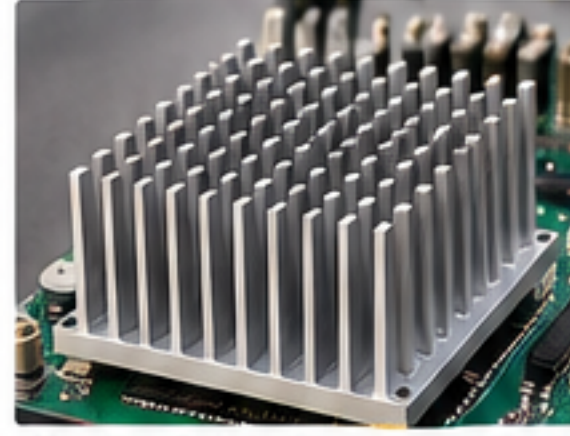
Plane Wall with Convection
on Both Sides

3 FINS / EXTENDED SURFACES

Fins combine conduction within the solid with convection from the surface.

3.1 Introduction to Fins

- **Purpose:** To increase the total surface area and enhance heat dissipation when the convection coefficient h is low.
- **Applications:** Heat sinks, radiators, engine cooling fins, electronics cooling.



Heat sink (electronics cooling)



Radiator (engine cooling)

3.2 General Fin Equation

Energy balance on a differential fin element yields:

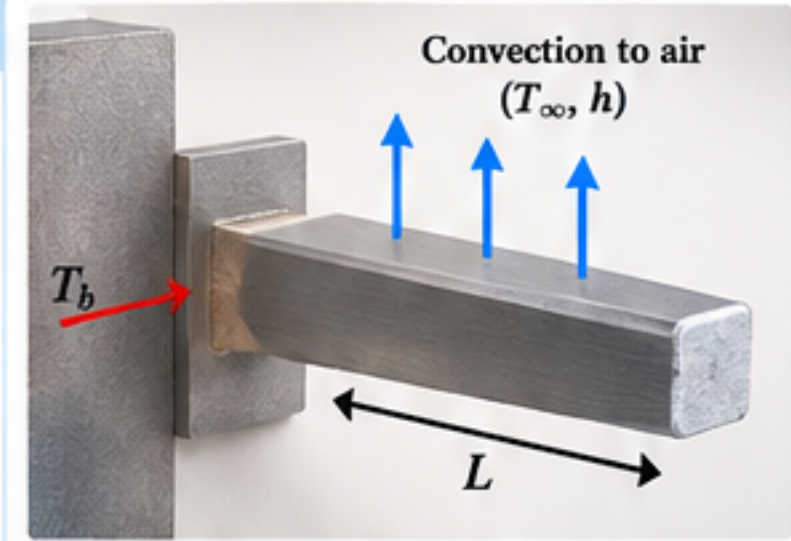
$$\frac{d^2\theta}{dx^2} - m^2\theta = 0 \quad (13)$$

Where: $\theta(x) = T(x) - T_\infty$, $m = \sqrt{\frac{hP}{kA_c}}$

(P = perimeter, A_c = cross-sectional area, k = thermal conductivity, h = convection coefficient).

General solution:

$$\theta(x) = C_1 \cosh(mx) + C_2 \sinh(mx) \quad (14)$$



Straight fin (extended surface)

3.3 Types of Fins — Solutions

Case 1: Fin with Insulated (Adiabatic) Tip

Boundary conditions: $\theta(0) = \theta_b = T_b - T_\infty$; $\left. \frac{d\theta}{dx} \right|_{x=L} = 0$

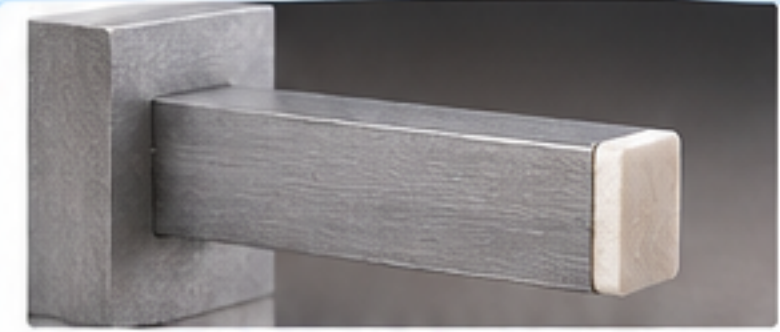
$$\frac{\theta(x)}{\theta_b} = \frac{\cosh[m(L-x)]}{\cosh(mL)} \quad (15)$$

$$q_f = \sqrt{hPkA_c} \theta_b \tanh(mL) = M \tanh(mL) \quad (16)$$

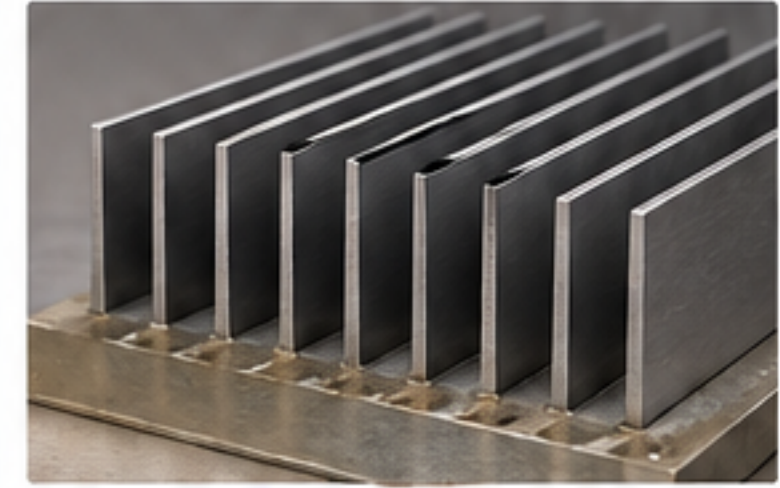
Case 2: Infinite Fin Approximation ($mL \gg 1$)

Boundary conditions: $\theta(0) = \theta_b$; $\theta(L \rightarrow \infty) = 0$

$$\frac{\theta(x)}{\theta_b} = e^{-mx}, \quad q_f = \sqrt{hPkA_c} \theta_b = M \quad (17)$$



Fin with insulated (adiabatic) tip



Infinite fin (long fin approximation)

3.4 Fin Performance Metrics

Fin Efficiency (η_f):

$$\eta_f = \frac{\text{Actual } q}{\text{Maximum possible } q} = \frac{q_f}{hA_f\theta_b} \quad (18)$$

- Adiabatic tip fin: $\eta_f = \frac{\tanh(mL)}{mL}$
- Infinite fin: $\eta_f \rightarrow \frac{1}{mL}$
- Circular pin fin (insulated tip): $\eta_f = \frac{\tanh(mL)}{mL}$, $m = \sqrt{\frac{4h}{kD}}$

Fin Effectiveness (ϵ_f):

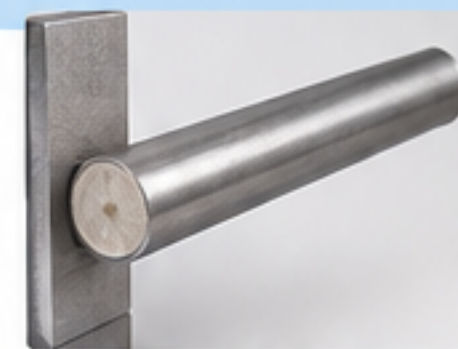
$$\epsilon_f = \frac{\text{Heat transfer with fin}}{\text{Heat transfer without fin}} = \frac{q_f}{hA_b\theta_b} \quad (19)$$

- Adiabatic tip fin: $\epsilon_f = \frac{\sqrt{hPkA_c} \tanh(mL)}{hA_c\theta_b}$
- Infinite fin: $\epsilon_f = \sqrt{\frac{kP}{hA_c}}$

Guideline: $\epsilon_f > 2$ is required to justify adding fins.

Overall Surface Efficiency (for N fins on total surface area A_t):

$$\eta_o = 1 - \frac{NA_f}{A_t} (1 - \eta_f), \quad q_{total} = \eta_o h A_t \theta_b \quad (20)$$



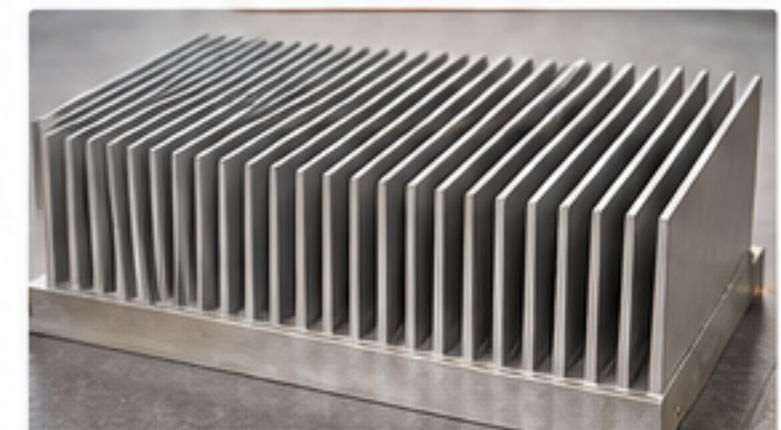
Circular pin fin (insulated tip)



Pin fin array (e.g., heat exchanger)



Finned heat exchanger (increased surface area)



Multiple fins on a surface (overall surface efficiency)

4 TRANSIENT CONDUCTION

4.1 Lumped Heat Capacitance Method

Assumes uniform internal temperature at any given instant: $T = T(t)$. Valid when internal conduction resistance is negligible compared to external convection resistance.

Biot Number Criterion:

$$\text{Bi} = \frac{hL_c}{k} < 0.1 \quad \text{where:} \quad L_c = \frac{V}{A_s} \quad (21)$$

Governing Equations:

$$-hA_s(T - T_\infty) = \rho V C_p \frac{dT}{dt} \quad (22)$$

$$\frac{T(t) - T_\infty}{T_i - T_\infty} = e^{-t/\tau} = e^{-\text{Bi} \cdot \text{Fo}} \quad \text{where:} \quad \tau = \frac{\rho V C_p}{h A_s}, \quad \text{Fo} = \frac{\alpha t}{L_c^2} \quad (23)$$

Total heat transferred up to time t :

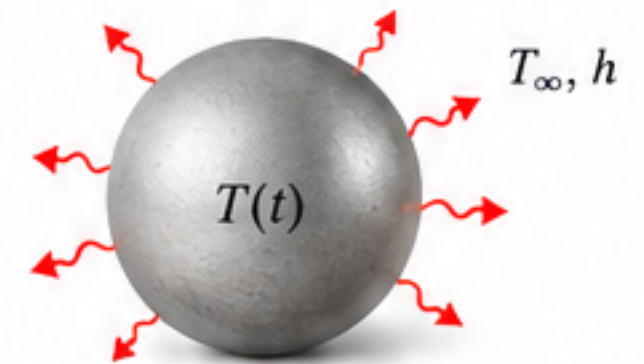
$$Q = \rho V C_p (T_i - T_\infty) \left(1 - e^{-t/\tau}\right) \quad (24)$$

Lumped Capacitance Example



A small metal sphere cooling in air (can be modeled using lumped heat capacitance when $\text{Bi} < 0.1$)

Heat Loss by Convection



Uniform internal temperature assumption

4.2 Semi-Infinite Solid Approximation

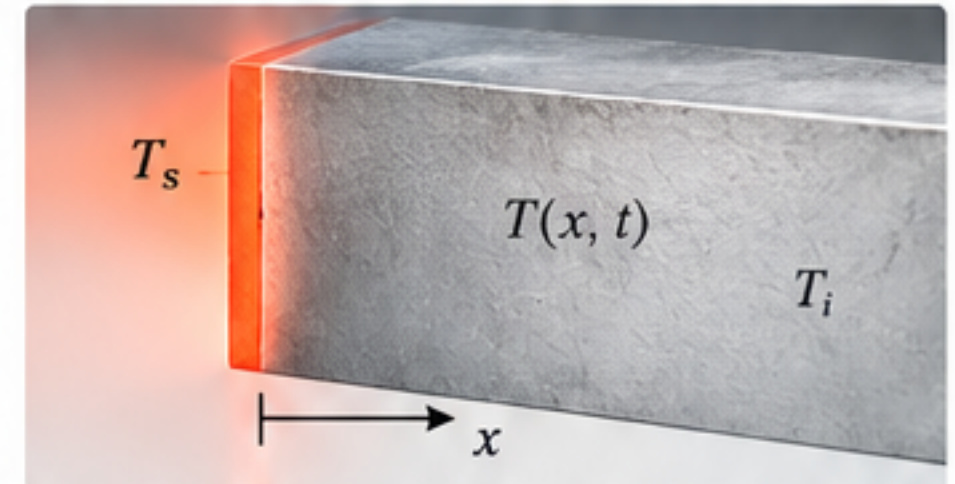
Used when thermal penetration depth is small relative to body thickness ($\delta \ll L$).

$$\text{Thermal penetration depth: } \delta \approx 4\sqrt{\alpha t} \quad (25)$$

Case 1: Constant Surface Temperature (T_s applied at $l = 0$)

$$\frac{T(x, t) - T_i}{T_s - T_i} = \text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right), \quad q_x''(t) = \frac{k(T_s - T_i)}{\sqrt{\pi \alpha t}} \quad (26)$$

Semi-Infinite Solid (Constant Surface Temperature)



Surface at constant temperature, heat penetrates into a semi-infinite solid

6 BOUNDARY LAYER THEORY

6.1 Hydrodynamic Boundary Layer

Flat plate, laminar flow (Blasius solution):

$$\delta(x) = \frac{5x}{\sqrt{\text{Re}_x}}, \quad C_{f,x} = \frac{0.664}{\sqrt{\text{Re}_x}}, \quad \bar{C}_f = \frac{1.328}{\sqrt{\text{Re}_L}} \quad (27)$$

Flat plate, turbulent flow:

$$\delta(x) \approx \frac{0.37x}{\text{Re}_x^{1/5}} \quad (28)$$

Velocity Boundary Layer over a Flat Plate



Development of hydrodynamic boundary layer on a flat plate

6.2 Thermal Boundary Layer

Local Nusselt number for laminar flat plate:

$$\text{Nu}_x = \frac{hx}{k} = 0.332 \text{Re}_x^{1/2} \text{Pr}^{1/3}, \quad \frac{\delta_t}{\delta} \approx \text{Pr}^{-1/3} \quad (29)$$

Thermal Boundary Layer



Thermal boundary layer (temperature profile) over a heated flat plate

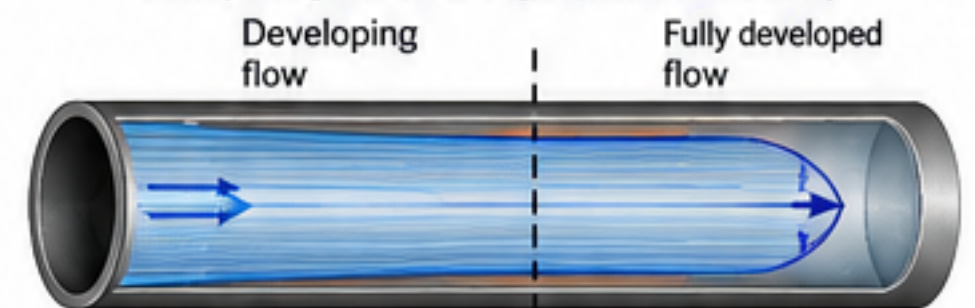
6.3 Entry Length in Internal Flow

$$\text{Laminar Hydrodynamic Entry Length: } x_{fd,h} \approx 0.05 \text{Re} D \quad (30)$$

$$\text{Laminar Thermal Entry Length: } x_{fd,t} \approx 0.05 \text{Re} \text{Pr} D \quad (31)$$

$$\text{Turbulent Entry Length (both): } x_{fd} \approx 10 D \quad (32)$$

Entry Region in a Pipe (Internal Flow)



Velocity profile development in a circular pipe

7 CONVECTIVE HEAT TRANSFER CORRELATIONS

7.1 Forced Convection — External Flow

Flat plate, laminar ($Re_L < 5 \times 10^5$):

$$\overline{Nu}_L = 0.664 Re_L^{1/2} Pr^{1/3} \quad (Pr \geq 0.6) \quad (33)$$

Flat plate, turbulent ($Re_L > 5 \times 10^5$):

$$\overline{Nu}_L = 0.037 Re_L^{4/5} Pr^{1/3} \quad (0.6 \leq Pr \leq 60) \quad (34)$$

Flat plate, mixed boundary layer:

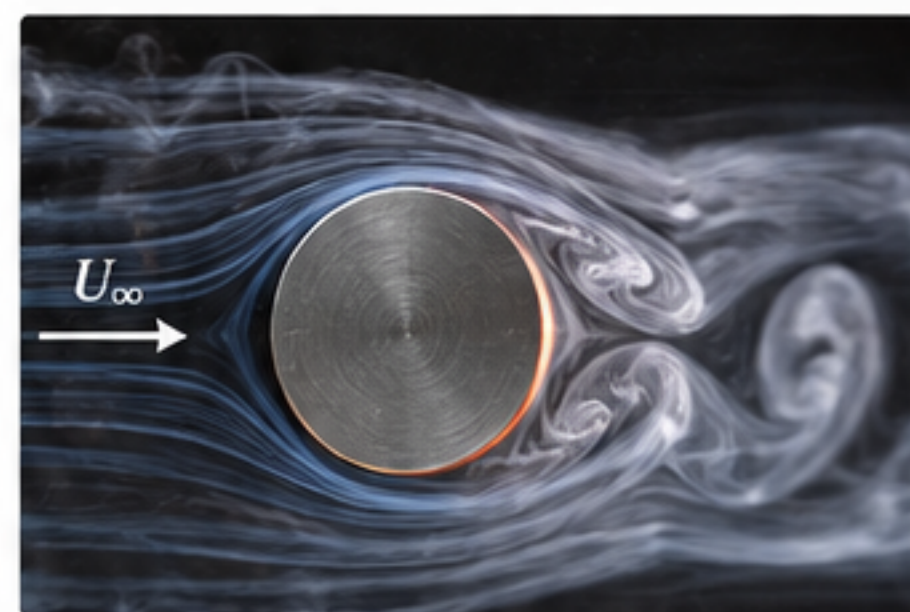
$$\overline{Nu}_L = (0.037 Re_L^{4/5} - 871) Pr^{1/3} \quad (35)$$

Flow over a cylinder (Churchill–Bernstein):

$$\overline{Nu}_D = 0.3 + \frac{0.62 Re_D^{1/2} Pr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}} \left[1 + \left(\frac{Re_D}{282000} \right)^{5/8} \right]^{4/5} \quad (36)$$



Flow over a flat plate
(external flow)



Flow over a cylinder
(vortex shedding in external flow)

7.2 Forced Convection — Internal Flow

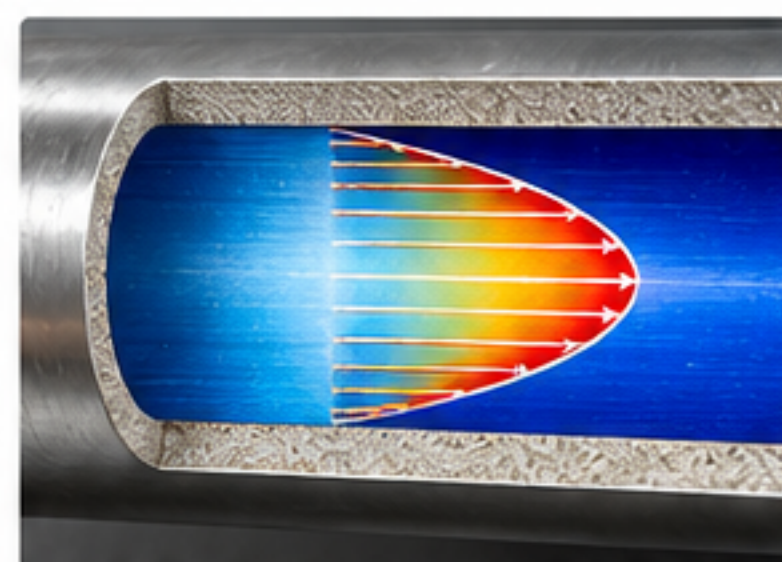
Fully developed laminar flow: $Nu_D = 3.66$ (constant T_s), $Nu_D = 4.36$ (constant q'').

Dittus-Boelter equation (turbulent flow):

$$Nu_D = 0.023 Re_D^{4/5} Pr^n \quad (n = 0.4 \text{ for heating, } n = 0.3 \text{ for cooling}) \quad (37)$$

Gnielinski correlation (smooth pipes, transitional/turbulent):

$$Nu_D = \frac{(f/8)(Re_D - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}, \quad f = (0.790 \ln Re_D - 1.64)^{-2} \quad (38)$$



Fully developed internal flow
in a circular pipe
(laminar velocity profile)

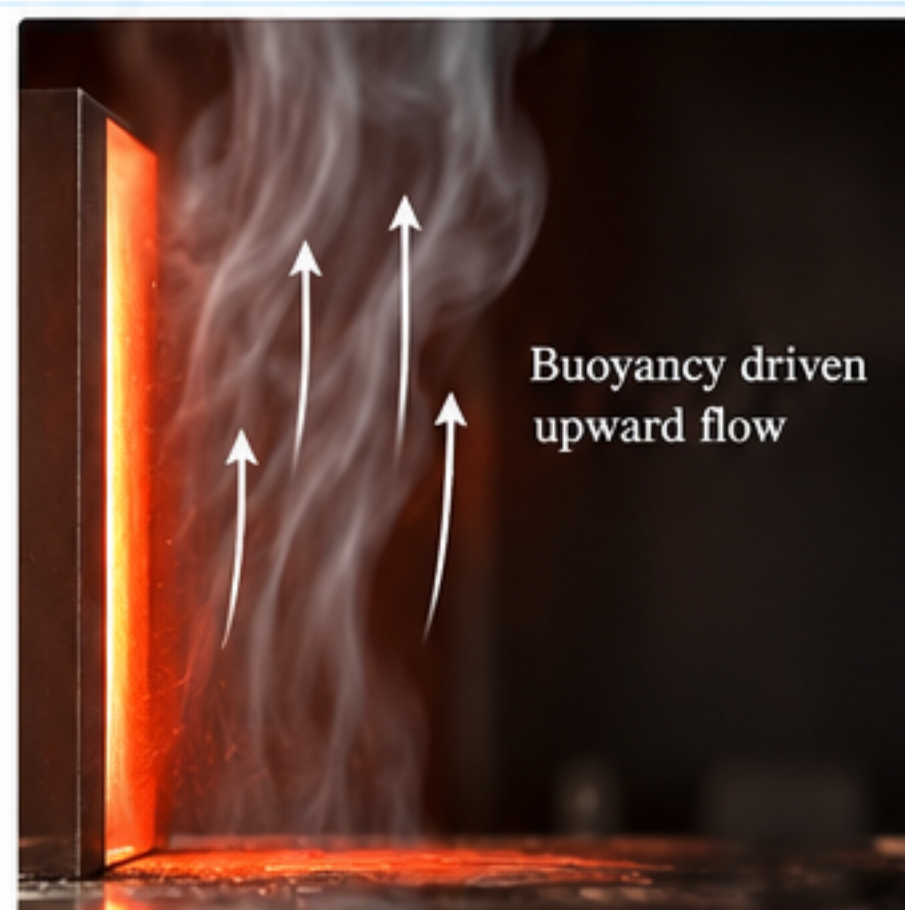
8 NATURAL CONVECTION

Volumetric thermal expansion coefficient:

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_p \quad \left(\text{For ideal gases: } \beta = \frac{1}{T_f} \right) \quad (39)$$

Vertical plate correlation (Churchill–Chu, all Ra_L):

$$\overline{Nu}_L = \left[0.825 + \frac{0.387 Ra_L^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}} \right]^2 \quad (40)$$



Natural convection from a vertical plate
(hot surface in still air)

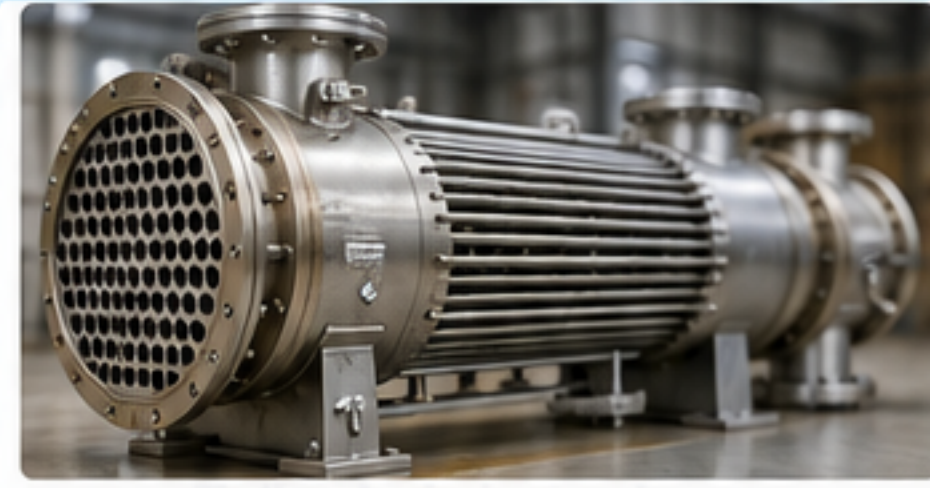
9 HEAT EXCHANGERS

Devices used to transfer heat between two fluids at different temperatures.

9.1 Energy Balance

$$q = \dot{m}_h c_{p,h} (T_{h,in} - T_{h,out}) = \dot{m}_c c_{p,c} (T_{c,out} - T_{c,in}) \quad (41)$$

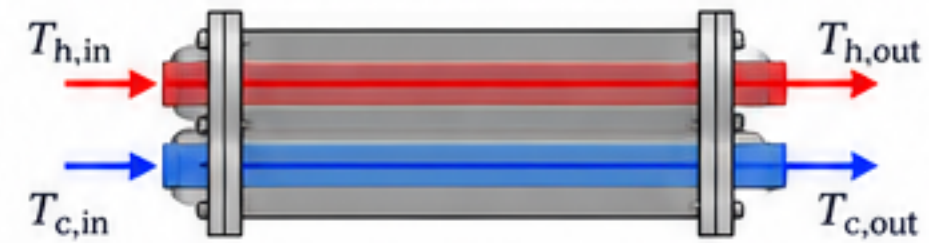
$$q = C_h (T_{h,in} - T_{h,out}) = C_c (T_{c,out} - T_{c,in}) \quad (42)$$



Shell-and-tube heat exchanger
(widely used in process industries)

9.2 LMTD Method

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}, \quad q = U A F \Delta T_{lm,cf} \quad (43)$$

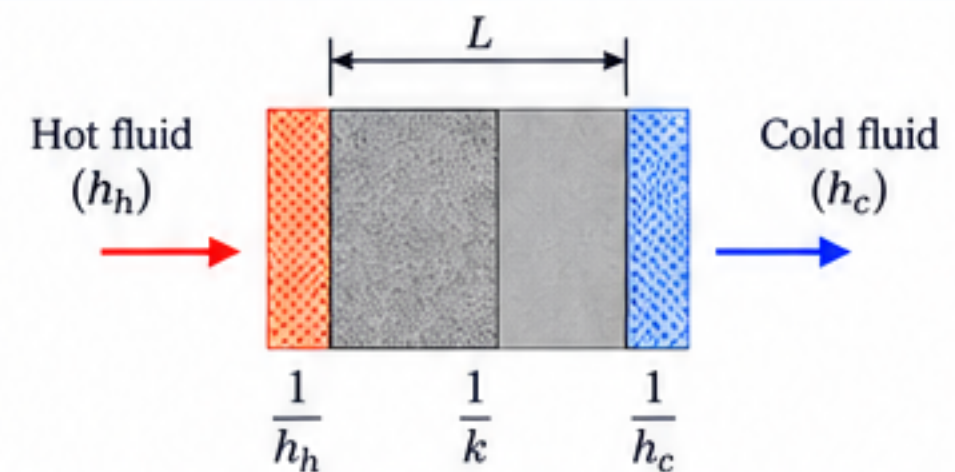


Counterflow heat exchanger
(hot and cold fluids flow in opposite directions)

9.3 Overall Heat Transfer Coefficient & Fouling

For a plain wall:

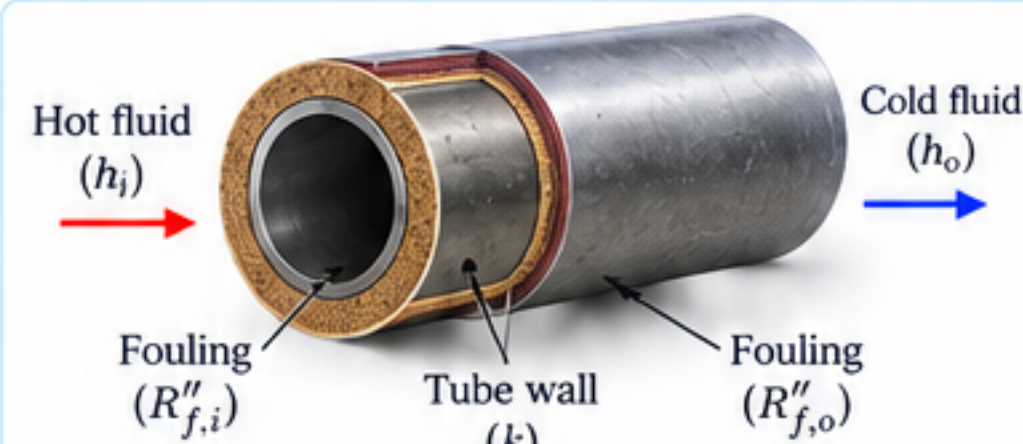
$$\frac{1}{U} = \frac{1}{h_h} + \frac{1}{k} + \frac{L}{k} + \frac{1}{h_c} \quad (44)$$



Heat transfer through a plain wall with convection on both sides

For a tubular wall with fouling factor R''_f :

$$\frac{1}{U_o A_o} = \frac{1}{h_i A_i} + \frac{R''_{f,i}}{A_i} + \frac{\ln(r_o/r_i)}{2\pi k L} + \frac{R''_{f,o}}{A_o} + \frac{1}{h_o A_o} \quad (45)$$



Tubular wall with fouling on inner and outer surfaces

9.4 Effectiveness-NTU Method

Number of Transfer Units (NTU):

$$NTU = \frac{UA}{C_{\min}}, \quad \varepsilon = \frac{q}{q_{\max}} = \frac{q}{C_{\min} (T_{h,in} - T_{c,in})} \quad (46)$$

Parallel flow:

$$\varepsilon = \frac{1 - \exp[-NTU(1 + C_r)]}{1 + C_r} \quad (47)$$

Counter flow ($C_r < 1$):

$$\varepsilon = \frac{1 - \exp[-NTU(1 - C_r)]}{1 - C_r \exp[-NTU(1 - C_r)]} \quad (48)$$

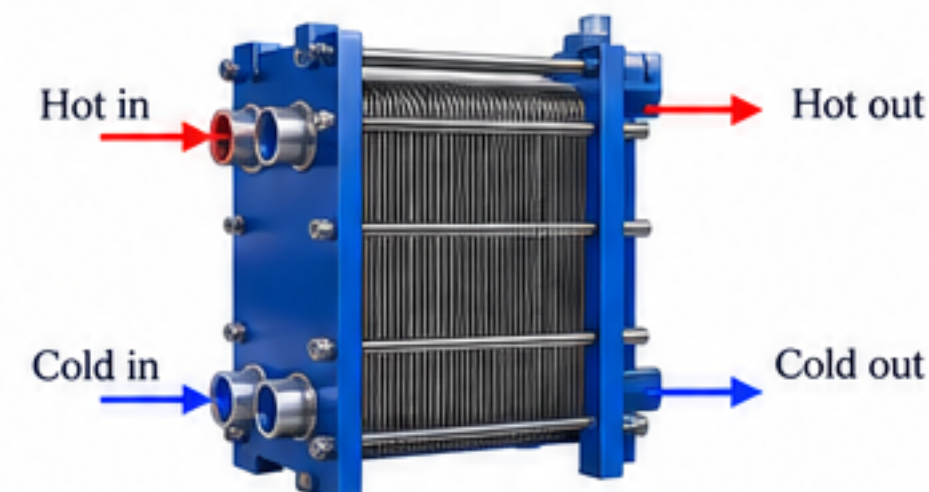


Plate heat exchanger
(compact and high effectiveness)

$$C_r = \frac{C_{\min}}{C_{\max}}, \quad 0 \leq C_r \leq 1$$

$$q_{\max} = C_{\min} (T_{h,in} - T_{c,in})$$



Previous Year Questions (PYQs):

https://drive.google.com/drive/folders/1YBFJfAqDRUcBhUY_BhzFKYsCi2Yg89Ew

Useful for practice and revision!